



The integration of share repurchases into investment decision-making: Evidence from Japan

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ABSTRACT

Share repurchases have become a significant global transaction and skepticism persists about their ambiguous effect on investment. We enter this debate motivated by inconclusive relevant empirical evidence and the remarkable evolution of share repurchases in Japan. Accordingly, we explore the investment-share repurchases nexus using panel data from Japanese listed firms between 2000 and 2019. We document a negative relationship consistent with the notion that firms curtail repurchases in the presence of investment spending and vice versa. We establish this contention by using subsamples stratified according to financing constraints, growth opportunities and cross-shareholdings. The findings endure after controlling for self-selection, endogeneity and heterogeneity and suggest that firms' investment decision precedes share repurchases which are adjusted accordingly. The results highlight how the widely criticized feature of cross-shareholdings, which is a distinct trait of the Japanese market, affects the integration of share repurchases into investment decision making.

1. Introduction

During the last two decades, corporate payout policy in Japan displayed a notable shift. As Fig. 1 shows, from 1991 until 2000 almost all of TOPIX firms distributed funds solely via dividends. However, this pattern has reversed gradually. Since 2001, on an annual basis, the majority of firms engage in share repurchases, and a dual payout policy prevails. This shift was facilitated through a series of policies and reforms which sought to address the severe downturn of the country's economy in the 1990s and the consequent prolonged stagnation. Distinct traits of corporate Japan, as bank-based practices, cross-shareholdings¹ and insular, stultifying corporate boards were recognized as leading to complacency, lower returns on equity and capital inefficiency. In this respect, the Japanese government considered the switch from traditional stakeholder-oriented to more shareholder-

oriented practices as vital for corporate Japan (Miyajima & Saito, 2019; Nakamura, 2011). Accordingly, the progressive deregulation² of share repurchases granted this distribution mechanism to Japanese firms. The aim was to provide firms with a flexible countermeasure to the long-lasting share price decline and to ease the dividend burden from equity issues (Kobayashi & Irome, 2012). Regarding the former, further downward pressure on share prices was expected due to the 'Act on Limitation on Shareholding by Banks and Other Financial Institutions' which aimed to the unwinding of cross-shareholdings. However, the rise of share repurchases initially and predominantly witnessed in the US and the UK has often been greeted with skepticism.

A sizable strand of the literature supports that share repurchases are considered a valuable tool due to their inherent flexibility. Contrary to the quasi-fixed cost nature of dividends, repurchase announcements³ are not legally binding, nor repurchase reductions lead to a decline in share

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¹ Cross-shareholdings are interconnected portfolios of ownership by listed Japanese companies in each other.

² See Kobayashi and Irome (2012) and Tong and Bremer (2016) for a more detailed presentation and discussion of changes surrounding share repurchases regulation.

³ This holds for open market share repurchases which are the most common repurchases method.

price (Allen & Michaely, 2003; Brav, Graham, Harvey, & Michaely, 2005; Campello, Graham, & Campbell, 2010; Lintner, 1956; Stephens & Weisbach, 1998). Thus, managers, as opposed to ‘sticky’ dividends, can with no hesitation restrain share repurchases to strengthen liquidity and/or to procure investment capital (Bliss, Cheng, & Denis, 2015; Brav et al., 2005; Iyer, Feng, & Rao, 2017). Share repurchases also represent a flexible method to distribute excess cash when there is a lack of attractive investment opportunities relative to the firm’s cash position, thus limiting the non-optimal use of excess cash by managers (Jensen, 1986; Tong & Bremer, 2016). This is usually referred to as the over-investment motive or hypothesis. Considering the positive impact of share repurchases on firms’ financial flexibility Bonaimé, Hankins, and Harford (2014) argue that share repurchases constitute a risk management device. The authors provide evidence that financial flexibility in payout decisions and hedging serving as substitutes for one another. On the other hand, several economists and policymakers find the popularity of share repurchases politically and economically untenable, fearing they may inhibit capital investments. The argument is that share repurchases can be used to generate short-term gains and to improve widely used performance and bonus-related metrics, i.e., Earnings Per Share (Almeida, Fos, & Kronlund, 2016; Kahle, 2002; Young & Yang, 2011). In contrast, capital expenditures represent a commitment, unappealing to managers in the face of uncertainty and/or stagnation. This may induce myopic business practices where managers may be willing to trade off investment for share repurchases, especially during economic downturns at times when investment is needed the most.⁴ Accordingly, evidence documents that increased share repurchase activity is capable of explaining the lack of translation of corporate profitability into growth in employment and overall economic prosperity (Lazonick, 2014). Companies that repurchase shares reduce their investment in employment, capital, and research and development (Turco, 2018). The progressive popularity of share repurchases in Japan generates tantamount concerns.

Thus, the objective of this study is to explore whether Japanese firms exploit the flexibility of share repurchases in managing their internal funds and thus foster capital investment. This is particularly significant since Japan represents one of the largest equity markets facing a prolonged period of economic stagnation. Moreover, Japanese firms due to their distinct traits are considered traditionally to be less reliant on internal funds. Specifically, stable long-term cross-shareholdings and firms’ close ties to their banks lead to a comparatively higher flow of information between the firm, its shareholders, and its lenders, lowering

asymmetric information and increasing the availability of external capital (Dewenter & Warther, 1998). However, since the 1990s and the beginning of a prolonged recession, bank financing was often problematic due to the unwinding of bank cross-shareholdings and the decline in the value of available collateral. Moreover, reforms under Abenomics targeted the poor capital efficiency of Japanese firms. Thus, our research objective will directly assess whether the deregulation of share repurchases benefited Japanese firms by granting them a valuable distribution mechanism which increased their financial flexibility. Although there is a number of studies on the phenomenon of share repurchases in the Japanese market our topic has not been previously covered. Earlier studies on Japan investigated the motives behind share repurchases (Tong & Bremer, 2016; Tong, Suzuki, Kato, & Bremer, 2012; Wang, 2020), the market’s reaction to share repurchases announcements (Zhang, 2002) and their relationship to the allocation of corporate control (Franks, Mayer, Hideaki, & Ogawa, 2018). These are surely interesting research topics yet different from ours. This study explores the investment-share repurchases nexus to assess whether Japanese firms exploit the flexibility of share repurchases in managing their internal funds.

We address the research objective discussed in the previous paragraph through several complementary estimation methods. First, as we are concerned with the over-time investment-repurchases relationship for any given firm, we regress investment on share repurchases and a set of control variables, restricting the variation only to within-firm. We document a negative relationship between shares repurchases and investment. For robustness, we also employ a variety of tests to address self-selection, endogeneity and heterogeneity and we use alternative proxies to verify the baseline model results. We test this relationship further with appropriately stratified subsamples. These estimations show a negative relationship that intensifies (attenuates) for firms with higher internal and external financing constraints, growth opportunities, and investment intensity. The results are consistent with the flexible use of share repurchases which allows firms to curb share repurchases in the presence of higher investment spending, financial constraints, and growth opportunities. Finally, we turn to quantile regressions to gain insight into the share repurchases-investment relationship into other parts of the investment distribution other than the conditional mean. Quantile regressions reveal an asymmetric negative relationship, becoming stronger at consecutive quantiles on the higher end of the investment distribution (75th, 90th, 95th). In line with the previous results, quantile regression estimation suggests that the flexi-

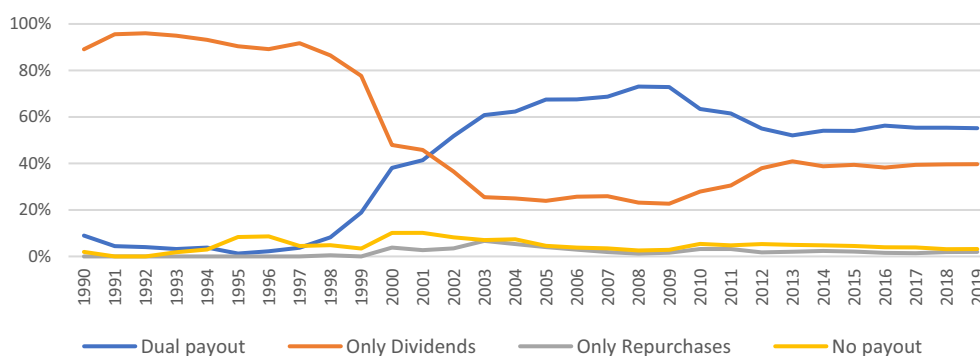


Fig. 1. The evolution of payout policy in Japanese listed firms, 1990–2019.
Notes: Sample is 23,740 firm-years observations from 1774 TOPIX firms.

⁴ This reasoning has led to a temporary ban on share repurchases for companies that received financial aid during the coronavirus pandemic in the US (see the CARES Act, 2020)

bility of share repurchases is more evident in firms with higher investment needs. However, it is important to note that the relationship between investments and share repurchases does not hold for firms with high company-cross shareholdings. This supports the wide criticism against this distinct feature of Japanese firms as it indicates that firms

who resist policy reforms do not improve their economic efficiency.

This study contributes to the literature by extending the investigation of the share repurchases- investments relationship in Japan. Since the majority of the relevant literature is US- and UK-focused, we directed this investigation into a country with marked differences in cultural and institutional characteristics. The findings suggest that share repurchases are integrated into corporate financial decision-making in Japan as a valuable tool due to their inherent flexibility. Japanese firms appear to mirror the practices of their Anglo-American counterparts as the findings are in line with firm-level empirical studies based on the US and the UK (Bliss et al., 2015; Chasiotis & Georgantopoulos, 2021; Iyer et al., 2017). This highlights the overarching need for financial flexibility and the importance of financing frictions in corporate decision-making. Moreover, the findings are consistent with managerial views that investment decisions are made first and repurchases are decided according to investment capital availability (Brav et al., 2005; PwC, 2019). Thus, share repurchases do not seem to suppress capital investment, thus, providing no support for the critique against this payout mechanism. Nevertheless, we also provide evidence that the widely criticized feature of cross-shareholdings in Japan affects the integration of share repurchases into investment decision making. These firms which resist policy reforms do not integrate share repurchases in a way that improves their financial flexibility. This supports plans of the Tokyo Stock Exchange to provide further incentives to Japanese firms to reduce their company cross-shareholdings. This study provides evidence directly related to the debate regarding the impact of share repurchases on investments, a matter of particular significance considering the poor economic efficiency of Japanese firms. Thus, it is expected to be of interest to economists, investors, and policymakers.

The rest of the paper is structured as follows. Section 2 reviews the extant literature on the capital investment-share repurchases relationship and develops hypotheses. Section 3 describes the sample and presents the research methodology. Section 4 presents and discusses the empirical results. Finally, Section 5 concludes.

2. Theory and hypotheses development

2.1. Motives for share repurchases

Since repurchase and dividends can be viewed as substitutes, certain theoretical arguments that address dividend policy can be applied to share repurchase as well, including the early signaling models (Miller & Rock, 1985; Ofer & Thakor, 1987). Thus, the signaling hypothesis implies that share repurchase is a signal of future over performance. A closely related theory is the undervaluation hypothesis. This is another version of the asymmetric information story. Managers have superior knowledge about their firm's true value and repurchase their shares when the market under values the firm. Under the validity of the undervaluation hypothesis, the literature assumes that there is informational asymmetry between a firm's executives and its investors. That is, the firm knows more about the true value of its stock than outside investors and can, therefore, capitalize on this informational advantage by repurchasing its stock at a price below its true value. The parallel between dividend theories and repurchase also applies to the diverse agency cost hypothesis. This research line has focused on repurchase as a solution to the over-investment problem (Jensen, 1986). These payouts reduce divertible cash flow and mitigate managements' incentives to invest in negative net present value projects (the over-investment hypothesis). There are also certain arguments that deal with the darker motives for repurchase (Kahle, 2002). It is possible that insiders can use

their valuable private information to time repurchase decisions to, say, maximize the value of their performance-based compensation or to engage in profitable insider trading. Another theoretical argument asserts that firms use stock options to compensate their employees with this practice increasing the number of shares outstanding, so the firm repurchases shares to maintain a relatively stable total share count (Brav et al., 2005). Finally, the literature has also explored the role of the evidence that in many country cases the personal income tax rate on capital gains is lower than the tax rate on dividends. This preferential tax hypothesis states that stock repurchases are preferred over dividends because the personal tax rate on capital gains is lower (Brav et al., 2005; Lie & Lie, 1999). In other words, when a firm has excess cash and decides to repurchase stock, there are no taxes paid by shareholders. Assuming zero dividends, investors can only experience gains in the form of price appreciation, or capital gains. Shareholders do not pay taxes on these gains until they realize the gains by selling their stock. Thus, when firms repurchase stock, shareholders have the freedom to make their own decision when to pay taxes on their gains. When a firm pays a dividend, shareholders do not have any control over the timing of their tax payments. Finally, share repurchases can also be used to provide a rapid capital structure adjustment towards the level, especially if they are followed by a cancellation of shares (Dhanani, 2016; Dittmar, 2000).

2.2. The relationship between investment and share repurchases

The relationship between investment and payouts was originally proposed by theoretical developments on asymmetric information and their impact on capital markets. Akerlof's (1970) seminal study on the market for 'lemons' provided insight on how asymmetric information between buyers and sellers can lead to market malfunctions. Cognizant of their informational handicap, buyers demand a risk-premium that reflects average-quality used cars. Extending this notion to financial markets, in the presence of asymmetric information external capital providers will undervalue high-quality, in terms of their growth opportunities and risk, firms. As a result, firms prefer to fund their investment opportunities with internal rather than external capital. This notion was subsequently embodied in the pecking order theory of financing which links firm investments with payouts. Firms prefer internal over external funds; however, the former may not always suffice (Myers, 1984; Myers & Majluf, 1984). Since cash flows are cyclical, firms may decrease dividend payments to avoid resorting to capital markets. Nevertheless, managers due to the 'stickiness' of dividends may be forced to seek external capital. In this case managers prefer debt over equity since the latter carries less asymmetric information costs. However, contrary to 'sticky' dividends, share repurchases due to their inherent flexibility (i.e., repurchases announcements are not legally binding and share repurchases reductions are not punished by the market via sharp price reductions) can be scaled back without hesitation in the presence of investment needs. Thus, helping the firm to avoid resorting to costly external finance. Nevertheless, Japanese firms which operate within a bank-based financial system are considered to suffer from less information asymmetries due to (i) stable long-term cross shareholdings and (ii) close ties to their banks (Dewenter & Warther, 1998).

Earlier research, around 1990–2000 suggests that this leads to a comparatively higher flow of information between the firm, its shareholders, and its lenders (see Dewenter & Warther, 1998). Therefore, the financing hierarchy may not be as relevant to Japanese firms, especially those with strong links to Japanese banks. In this respect, Hoshi, Kashyap, and Scharfstein (1991) find that for such firms, investment is

less sensitive to liquidity. Nevertheless, since the burst of the Japanese 'bubble economy' in the 1990s these distinct traits of the Japanese economy are undergoing a rapid change. Firms become less reliant on bank financing which became problematic as the value of available collateral declined (Yoshikawa & Gedajlovic, 2002). Moreover, market downturns hampered banks' capital levels through their market equity portfolios. Accordingly, the Japanese government and the Bank of Japan through a series of measures and efforts managed to reduce the overall level of bank shareholdings from 30% of the market capitalization in the 1990s to almost 10% in 2018 (Suga, 2018). Moreover, further changes on the governance code aimed at reducing company cross-shareholdings for two reasons. The first is that they lock shareholders equity preventing it from being redeployed elsewhere at higher returns. The second is that cross-shareholders lend each other unearned support thus leading to complacent, under-performing management which ignores shareholder's interests. These changes should lead Japanese firms to become more reliant on internal funds and more efficient in their management. From this perspective, share repurchases due to their inherent flexibility may prove as a useful tool in coordinating investment and payouts.

The flexible nature of share repurchases is consistently supported by US and UK focused managerial surveys. Managers underline the rigid of nature dividends and the contrasting flexibility of share repurchases. (Brav et al., 2005; Dhanani, 2016). Moreover, managerial views are consistent that investment decisions are made first and repurchases are decided according to investment capital availability (Brav et al., 2005; PwC, 2019). This suggests a negative relationship between share repurchases and capital investment as share repurchases can be scaled back with no hesitation in the presence of investment needs and vice versa. This is consistent with empirical evidence from US and UK firms that document a negative relationship between share repurchases and investment (Chasiotis & Georgantopoulos, 2021; Iyer et al., 2017). Considering the discussion in this paragraph, we formulate the following hypothesis:

H1. There is an inverse relationship between investment and share repurchases.

All else equal, the inverse relationship between investment and share repurchases should be more apparent in firms that are financially constrained. Firms unable to generate sufficient internal funds are expected to curtail repurchases more to secure sufficient investment capital. Moreover, external capital for financially constrained firms is likely to be comparatively costlier (Campello et al., 2010). Bliss et al. (2015) monitor firm behavior through the 2008–09 credit crisis and document large reductions in the share repurchases of US firms directed to fund investment. Also, supportive are findings by Iyer et al. (2017) and Chasiotis and Georgantopoulos (2021) which report a stronger negative share repurchases-investment relationship for financially constrained the US and UK listed firms. Thus, we hypothesize that for Japanese listed firms:

H2. Financial constraints attenuate the inverse relationship between investment and share repurchases.

Finally, the flexibility of share repurchases should be of particular importance to firms with, particularly high growth opportunities and investment spending (investment intensity). Ceteris paribus, firms with higher investment opportunities/intensity are expected to reduce share repurchases comparatively more to avoid the higher costs of external finance. Thus, we hypothesize that:

H3. Growth opportunities and investment intensity attenuate the inverse relationship between investment and share repurchases.

Corporate governance reforms under Abenomics aim to reduce company cross-shareholdings in Japanese (Miyajima & Saito, 2019). However, some firms are resisting this change as while some decrease their cross-shareholdings others are raising them. For the latter, Franks

et al. (2018) provide evidence that share repurchases are used by entrenched management as a method to preserve cross-shareholdings. Specifically, when shares are bought back from insiders who want to cash out, the firm can resell these shares to other insiders or strategic partners. This 'round trip' maintains high cross-shareholdings and ensures that shares do not end up to unwanted outsiders keeping management isolated from market pressure while maintaining unearned support from cross-shareholders. Accordingly, Miyajima and Ogawa (2016) point out that foreign investors antagonize entrenched management. In this respect, the Tokyo Stock Exchange is planning to reorganize its listings in 2022 into three groups: prime, standard and growth.⁵ As a further motive, membership to the sought-after prime index will depend on each company's March 31 level of free-floating market capitalization excluding cross-held shares. Firms that maintain high cross-shareholdings indicate poor economic efficiency. Therefore, we expect these firms not to exploit the flexible nature of share repurchases and thus not integrate them into their investment decision making. In contrast, we would expect the opposite for firms with low cross-shareholdings whose management lacks the supportive cushion of investor support and thus must be efficient with their capital. Specifically, we expect an insignificant relationship between investment and share repurchases for firms with high cross-shareholdings and a negative relationship for firms with low cross-shareholdings.

H4a. There is no relationship between investment and share repurchases for firms with high cross-shareholdings.

H4b. There is an inverse relationship between investment and share repurchases for firms with low cross-shareholdings.

3. Sample and methodology

3.1. Sample

To construct our sample, we retrieve information from Thomson Reuters' DataStream database and collect annual data for Tokyo Stock Price Index (TOPIX) firms. TOPIX tracks all domestic companies listed in the First Section of the Tokyo Stock Exchange. We choose 2000–2019 as our sample period since Japanese firms did not practically engage in share repurchases before 2000. We exclude from our sample financial and utility firms as well as firms with missing observations. The final sample consists of an unbalanced panel of 23,740 firm-year observations, from 1774 firms. We keep our panel in its unbalanced form to mitigate selection and survivorship bias. To reduce the potential impact of outliers we winsorize⁶ our variables at the 1st and 99th percentiles.

3.2. Baseline model

As our objective is to investigate the relationship between investment and share repurchases, we regress⁷ investment on share repurchases and a vector of control variables. As we are interested in the over-time investment-repurchases relationship for any given firm we use fixed effects at the firm level restricting the variation to within the firm. Thus, our baseline model for testing hypothesis H₁ is:

⁵ Japan Inc. faces a potential forced sell-off of cross-shareholdings. Financial times. (January. 20,2021), available at: <https://www.ft.com/content/e027bb9e-a57d-4e80-b444-70e13f68cdaa>. Accessed (20/4/2021)

⁶ Winsorisation is not applied to our variables when quantile regressions are applied since this technique is robust to the presence of outliers.

⁷ Our approach is similar to Iyer et al. (2017) and Chasiotis and Georgantopoulos (2021).

$$Investment_{i,t} = \alpha_0 + \beta_1 Repurchases_{i,t-1} + \beta_2 Controls_{i,t} + firm_i + year_t + \varepsilon_{i,t} \quad (1)$$

$Investment_{i,t}$ is the investment variable and $Repurchases_{i,t-1}$ the lagged⁸ share repurchases variable. $Controls$ is a set of control variables; α_0 is the constant; $firm$ controls for individual firm-effects; $year$ is a year-dummy controlling for time-series trends and u is the disturbance term. The subscript i denotes firms and the subscript t denotes years. According to hypothesis H_1 we expect β_1 to be negative and statistically significant.

Investment (*Investment*) and share repurchases (*Repurchases*) are measured as capital expenditures and share repurchases scaled by total assets, respectively. Our set of control variables is drawn from extant literature on investment (Aivazian, Ge, & Qiu, 2005; Baskin, 1989; Bond & Meghir, 1994; Carpenter & Guariglia, 2008; Fazzari, Hubbard, Petersen, Blinder, & Poterba, 1988; Iyer et al., 2017). Specifically, we control for a firm's operating cash flows (*Cash Flows*) and tangibility (*Tangibility*). These variables are expected to be positively related to investment as they are related to the firm's ability to generate funds and to borrow external capital, respectively. We control for dividends (*Dividends*) as a significant use of funds competing to investment. Firm size (*Firm Size*), is controlled for as larger firms may grow at a slower rate. Moreover, firms with higher growth opportunities (*Growth*) are expected to invest more. In addition, we control for leverage (*Leverage*) as high levels of debt obligations may lead to liquidity problems. Finally, following Iyer et al. (2017) and Gugler (2003), we include firm age (*Firm age*) and depreciation (*Depreciation*). To test hypotheses H_2 , we first measure internal and external financing constraints according to conventional measurements used in the literature (Shyam-Sunder & Myers, 1999; Frank & Goyal, 2003; Baños-Caballero, García-Teruel, & Martínez-Solano, 2014; Byoun & Xu, 2016, see Table A1, appendix for variable definitions). Following Myers (2001) we emphasize the share repurchases-investment relationship under several firm-specific characteristics.

Accordingly, we stratify firms into the relevant subsample depending on whether their measurement lies above or below the sample's median. For hypotheses H_3 and $H_{4a,b}$ we stratify our firms into high/low growth opportunities and company cross-shareholdings subsamples according to their growth opportunities and company cross-shareholdings variables. According to hypotheses H_2 , and H_3 the coefficient of share repurchases (β_1) is expected to be strongly negative in the subsamples where firms are characterized by higher financial constraints, growth opportunities. According to hypotheses H_{4a} and H_{4b} the coefficient of share repurchases (β_1) is expected to be insignificant in the subsample of firms with high cross-shareholdings, and negative at the subsample of firms with low cross-shareholdings. Variable definitions and summary statistics are provided in Table A1, Appendix and Table 1, respectively.

The average (median) investment for our sample is 3,7% (2,9%). The dividend ratio ranges from 0 to 5,1% with an average value of 1%. The respective share repurchases ratio ranges from 0 to 6,9% with an average value of 0,4%. The mean (median) firm has a 7,2% (6,7%) cash ratio while 29% (27,5%) of its assets are tangible. The average (median) leverage ratio of the firms in our sample is 48,3% (48,4%). Finally, the average (median) Growth is 1,55 (1,08). In terms of size, the mean (median) firm in our sample has a book value of total assets equal to 99 billion ¥ (85 billion ¥). Table A2 (Appendix) presents the pairwise correlation matrix. The correlation between investment and share repurchases is negative and statistically significant as expected. The same is observed between dividends and investment. The correlation coefficients between investment and all control variables are positive and statistically significant. This is unexpected in the case of size as larger firms are often theorized to grow at a slower pace. However,

further factors need to be controlled for as said correlations may be spurious thus further factors are controlled for in our regression analysis. and leverage is positive. To test for the presence of multicollinearity in our model we compute of variance inflation factors (see Table A3, Appendix). Table A3 in the Appendix indicates that multicollinearity is not an issue. The general rule of thumb is that multicollinearity does not represent an issue if VIF are smaller than 10 (Gujarati, 2004).

3.3. Robustness checks

To test the robustness of our results we employ a series of alternate methodologies. First, we employ the unconditional quantile fixed effects estimator by Borgen (2016) to address and investigate heterogeneous firm behavior in financial decision-making as often reported in the literature (Fattouh, Scaramozzino, & Harris, 2005, 2008; Sánchez-Vidal, 2014; Mathur, Rao, Strain, & Veuger, 2016). Myers (1984) suggests that firms with insufficient cash flows, may reduce payouts to procure investment capital. However, ceteris paribus, this notion is expected to be relevant (irrelevant) for firms with higher (lower) investment activity. Therefore, quantile regressions allow us to examine whether the share repurchases-investment relationship varies between firms with different investment levels. We choose Borgen's⁹ (2016) extension of unconditional quantile regression as contrary to conditional quantile regressions the respective estimation is not influenced by the choice of regressors. UCR estimation uses the Recentered Influence Function (RIF). The RIF of the τ^{th} quantile of the dependent variable's distribution is defined by Firpo, Fortin, and Lemieux (2009) as:

$$RIF(Y_{it}; q_\tau, F_{Y_{it}}) = q_\tau + \frac{\tau - 1\{Y_{it} \leq q_\tau\}}{f_Y(q_\tau)} \quad (2)$$

where Y_{it} is the dependent variable, q_τ is the value of the dependent variable at the τ^{th} quantile, $F_{Y_{it}}$ is the cumulative distribution function of Y_{it} , $1\{Y_{it} \leq q_\tau\}$ is an dummy variable indicating whether the dependent variable is below q_τ and $f_Y(q_\tau)$ is the density at q_τ .

Furthermore, we address the selection-bias that may affect the validity of our OLS findings by applying the two-stage Heckman (1979) model. The first stage of the model estimates the probability of repurchases by using a probit regression, while the second stage corrects for potential self-selection by reforming and incorporating in the estimation the individual predicted probabilities. The selection equation has the following form:

$$DI_{i,t}^* = \gamma Z_{i,t} + \varepsilon_{i,t} \quad (3)$$

$$\text{where : } DI_{i,t} = \begin{cases} 1, & \text{if } Repurchases_{i,t}^* > 0 \\ 0, & \text{if } Repurchases_{i,t}^* = 0 \end{cases}$$

where $DI_{i,t}^*$ is a dummy latent variable that captures share repurchase activity (specification 1) and share repurchase increase (specification 2), γ is a vector of the coefficients to be estimated and $Z_{i,t}$ is a set of explanatory variables of $DI_{i,t}$ drawn from the extant literature and $\varepsilon_{i,t}$ is the error term. More specifically, as explanatory variables we include well-known share repurchases determinants as cash flows (*Cash Flows*), firm risk (*Stock's Beta*), firm size (*Firm Size*), stock options (*Stock Options*), dividends (*Dividends*), leverage (*Leverage*), growth opportunities (*Growth*), stock return (*Stock returns*), ownership concentration (*Ownership concentration*) and company cross-shareholdings (*cross-shareholdings*), (Almeida et al., 2016; Andriosopoulos & Hoque, 2013; Bens, Nagar, Skinner, & Wong, 2003; Dittmar, 2000; Franks et al., 2018;

⁸ Share repurchases and dividends (one of the control variables) are lagged for one year to mitigate endogeneity concerns.

⁹ For a comprehensive discussion between unconditional and conditional quantile regressions and relevant applications see Koenker and Bassett (1978), Firpo et al. (2009), Borah and Basu (2013), Markowitz, Adams, Lewitt, and Dunlop (2017) and Ma, Renwick, and Greig (2019).

Table 1

This table provides descriptive statistics for all the variables used in our models. Variable definitions are provided in Table A1, appendix.

	N	Mean	SD	p25	Median	p75	Min	Max	Skewness	Kurtosis
Investment	23,740	0.037	0.032	0.013	0.029	0.051	0.001	0.164	1.544	5.732
Repurchases	23,740	0.004	0.011	0.000	0.000	0.001	0.000	0.069	4.099	21.412
Dividends	23,740	0.010	0.009	0.004	0.008	0.012	0.000	0.051	2.168	9.070
Leverage	23,740	0.483	0.198	0.329	0.484	0.632	0.092	0.904	0.035	2.194
Leverage'	23,740	0.097	0.106	0.007	0.064	0.153	0.000	0.462	1.308	4.354
Cash flows	23,740	0.072	0.049	0.041	0.067	0.097	-0.052	0.234	0.587	4.229
Cash flows'	23,740	0.093	0.062	0.056	0.087	0.124	-0.076	0.299	0.560	4.518
Tangibility	23,740	0.290	0.169	0.169	0.275	0.391	0.008	0.763	0.515	2.954
Growth	23,740	1.549	1.532	0.730	1.080	1.720	0.320	10.310	3.394	17.019
Growth'	23,740	0.043	0.125	-0.019	0.033	0.094	-0.284	0.569	0.996	6.727
Firm size	23,740	18.354	1.492	17.375	18.200	19.218	14.986	22.491	0.403	3.145
Firm size'	23,740	17.681	1.305	16.579	17.472	18.682	15.938	19.965	0.378	1.947
Stock options	23,740	0.000	0.000	0.000	0.000	0.000	0.000	0.003	7.840	65.037
Ownership concentration	23,740	0.366	0.179	0.231	0.358	0.493	0.022	0.781	0.174	2.344
Stock returns	23,740	0.052	0.323	-0.146	0.048	0.252	-0.785	0.941	0.056	3.297
Cross shareholdings	23,740	0.142	0.165	0.000	0.080	0.230	0.000	0.960	1.238	3.940
Firm age	23,740	3.927	0.659	3.761	4.127	4.304	0.693	4.934	-2.143	8.799
Depreciation	23,740	0.033	0.023	0.018	0.030	0.044	-0.044	0.542	2.539	26.762

Jensen, 1986; Lee & Suh, 2011; Oswald & Young, 2008; Rozeff, 1982; Tong et al., 2012; Tong & Bremer, 2016; Wang, 2020) Variable definitions are provided in Table A1, appendix. To account for self-selection bias, we construct Eqs. (4) and (5) for repurchasers and non-repurchasers respectively.

$$E[Investment | DI_{i,t} = 1] = \beta'X + \delta + E[e | DI_{i,t} = 1] \\ = \beta'X + \delta + \rho\sigma_e \frac{\varphi(\omega'A)}{\Phi(\omega'A)} \quad (4)$$

$$E[Investment | DI_{i,t} = 0] = \beta'X + \rho\sigma_e \frac{-\varphi(\omega'A)}{1 - \Phi(\omega'A)} \quad (5)$$

To extract the expected impact of repurchase activity on the level of *Investment*, we subtract Eq. (5) from Eq. (4):

$$E[Investment | DI_{i,t} = 1] - E[Investment | DI_{i,t} = 0] = \delta + \rho\sigma_e \frac{\varphi(\omega'A)}{\Phi(\omega'A)(1 - \Phi(\omega'A))} \quad (6)$$

where ω' is a vector of coefficients to be estimated, φ stands for the distribution function of the standard normal distribution and Φ stands for the distribution function of the cumulative distribution function. Eq. (6) provides information on the effect of repurchases activity on *Investment* via the δ coefficient, which is related to the b_1 coefficient of Eq. (1). The potential selection bias can be traced and eliminated through the Inverse Mills ratio (*IMR*), which is a correction term on the condition that *repurchase activity* is positive. *IMR* is calculated as follows:

$$IMR = \frac{\varphi(\omega'A)}{\Phi(\omega'A)} \text{ if } DI_{i,t} = 1 \text{ or } IMR = \frac{-\varphi(\omega'A)}{1 - \Phi(\omega'A)} \text{ if } DI_{i,t} = 0 \quad (7)$$

We also address the endogeneity concern in financial decision making¹⁰ with the following estimation techniques. Initially, we use two stage least squares estimations (2SLS). As an additional technique, we employ entropy balancing to address potential endogeneity (see Arifin, Hasan, & Kabir, 2020; Ashraf, Michas, & Russomanno, 2019; Hainmueller, 2012; Hasan et al. 2021). Entropy balancing is a powerful method that leads to improved covariate balance. Moreover, it reduces loss of information as it does not 'match or discard' each unit. Finally, as additional robustness checks, we use alternative measures for financial constraints and for several explanatory variables and re-estimate our baseline model using the full sample and subsamples.

4. Empirical results

4.1. Baseline model results

Table 2 (Panel A) presents results from high-dimensional fixed-effects regressions for the full sample as well as several subsamples stratified according to high/low internal and external financial constraints, growth opportunities and cross-shareholdings. In Table 2 (Panel B) we follow Iyer et al. (2017) and report results using the same approach as in panel A but for additional robustness we restrict our analysis to repurchasing firms (firms that repurchase and pay dividends, and firms that only repurchase). By doing so we ensure that our results are not affected: i) from the number of firms which do not repurchase, ii) by confounding effects that may arise from fundamental differences between dividend-only or zero payout firms and those with repurchase activity. The full sample (Panel A) and the restricted sample (Panel B) document very similar findings. In the following part we focus our discussion on the full sample (Panel A).

The coefficient of the share repurchases variable (*Repurchases*) is negative and statistically significant for H_1 and is in line with managerial contentions regarding the flexible use of share repurchases (Brav et al., 2005; PwC, 2019). Specifically, this inverse relationship is consistent with the notion that share repurchases are easily restrained in the presence of investment spending and vice versa. Similar findings are reported by Iyer et al. (2017) and Chasiotis and Georgantopoulos (2021) for US and UK firms, respectively.

Table 2 also shows the coefficient of share repurchases in various subsamples of firms with high/low levels of internal and external constraints and growth opportunities. The repurchases coefficient is negative and statistically significant for both the low and high internal financial constraint samples. However, its size increases fivefold for the latter indicating that the inverse relationship between share repurchases and investment is more profound for firms that produce less internal funds compared to their investment needs. This observation holds when creating subsamples according to external financial constraints. For firms with low external constraints the share repurchases coefficient is statistically insignificant. This suggests that firms with access to cheap external finance do not need to coordinate share repurchase activity to their investment spending. On the contrary, for the high external constraints subsample, the share repurchases coefficient is negative, statistically significant, and threefold in size. Thus, results show clear support to hypothesis H_2 . Moreover, looking at Table 2 and results regarding subsamples stratified according to growth opportunities we can find support for hypotheses H_3 . The relationship between share repurchases and investment is statistically insignificant for firms with

¹⁰ See McCabe (1979), Crutchley, Jensen, Jahera, and Raymond (1999), Aggarwal and Kyaw (2010) and Lee, Liang, and Lin (2016).

Table 2

Baseline results from estimating Eq. (1) with high dimensional firm and year fixed effects. Panel A includes all firms in terms of payout policy (firms that pay dividends and repurchase, firms that pay only dividends, firms that only repurchase, and zero-payout firms). Panel B includes only repurchasing firms (firms that pay dividends and repurchase, and firms that only repurchase). In each panel the full sample is split into subsamples based on financing constraints, growth opportunities and cross-shareholdings. Robust standard errors clustered at the firm level are reported in parentheses. Variable definitions are reported in Table A1, appendix. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Full sample	Subsamples based on financing constraints, growth opportunities and cross-shareholdings							
		Internal Financing Constraints		External Financing Constraints		Growth opportunities		Cross-Shareholdings	
		Low	High	Low	High	Low	High	Low	High
Panel A									
Repurchases	− 0.037** (0.015)	− 0.017* (0.009)	− 0.084*** (0.026)	− 0.023 (0.017)	− 0.060** (0.025)	− 0.019 (0.027)	− 0.041** (0.018)	− 0.052** (0.020)	− 0.016 (0.022)
Dividends	0.186*** (0.050)	0.042 (0.028)	0.362*** (0.088)	0.084 (0.054)	0.433*** (0.086)	0.368*** (0.085)	0.185*** (0.061)	0.228*** (0.068)	0.187*** (0.069)
Leverage	− 0.014*** (0.003)	− 0.011*** (0.002)	− 0.012*** (0.005)	0.007 (0.006)	− 0.020*** (0.005)	− 0.010** (0.005)	− 0.016*** (0.005)	− 0.008* (0.004)	− 0.018*** (0.005)
Firm size	0.003** (0.001)	0.000 (0.001)	0.002 (0.001)	0.002 (0.001)	0.003** (0.002)	0.007*** (0.002)	0.002 (0.001)	0.003** (0.002)	0.004** (0.001)
Tangibility	0.138*** (0.007)	0.030*** (0.003)	0.156*** (0.008)	0.174*** (0.012)	0.123*** (0.008)	0.125*** (0.008)	0.159*** (0.010)	0.161*** (0.010)	0.132*** (0.009)
Growth	0.002*** (0.000)	0.001*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.004*** (0.002)	0.001*** (0.000)	0.002*** (0.000)	0.001*** (0.000)
Cash Flows	0.026*** (0.007)	0.064*** (0.005)	0.140*** (0.011)	0.026*** (0.008)	0.020* (0.011)	0.000 (0.011)	0.023*** (0.008)	0.027*** (0.010)	0.020** (0.010)
Firm Age	− 0.010*** (0.003)	− 0.002** (0.001)	− 0.014*** (0.004)	− 0.006** (0.003)	− 0.011** (0.005)	− 0.005** (0.003)	− 0.011*** (0.003)	− 0.003 (0.003)	− 0.018*** (0.004)
Depreciation	0.042 (0.032)	0.123*** (0.017)	0.014 (0.039)	0.022 (0.040)	0.039 (0.052)	0.022 (0.044)	0.038 (0.034)	− 0.022 (0.040)	0.081* (0.045)
Observations	23,740	11,648	11,810	11,908	11,626	11,956	11,566	11,791	11,751
Firm fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
R ²	0.666	54.769	35.280	45.064	12.643	12.913	7.117	30.185	43.613
Panel B									
Repurchases	− 0.044** (0.018)	− 0.020* (0.011)	− 0.073** (0.029)	− 0.027 (0.021)	− 0.056** (0.027)	− 0.021 (0.031)	− 0.042* (0.022)	− 0.061** (0.026)	− 0.022 (0.027)
Dividends	0.150** (0.059)	0.013 (0.033)	0.445*** (0.104)	0.104 (0.071)	0.332*** (0.098)	0.316*** (0.096)	0.120* (0.071)	0.207** (0.090)	0.124 (0.077)
Leverage	− 0.005 (0.004)	− 0.008*** (0.003)	− 0.001 (0.006)	0.015** (0.008)	− 0.010* (0.005)	− 0.004 (0.006)	− 0.007 (0.006)	− 0.001 (0.005)	− 0.009 (0.006)
Firm size	0.005*** (0.001)	0.002* (0.001)	0.004* (0.002)	0.004* (0.002)	0.006*** (0.002)	0.008*** (0.002)	0.005** (0.002)	0.007*** (0.002)	0.005*** (0.002)
Tangibility	0.151*** (0.009)	0.035*** (0.005)	0.170*** (0.012)	0.173*** (0.016)	0.145*** (0.011)	0.142*** (0.011)	0.168*** (0.012)	0.165*** (0.012)	0.150*** (0.013)
Growth	0.001*** (0.000)	0.001*** (0.000)	0.003*** (0.001)	0.001*** (0.000)	0.001** (0.001)	0.004** (0.002)	0.001** (0.000)	0.001** (0.001)	0.001*** (0.000)
Cash Flows	0.033*** (0.010)	0.067*** (0.007)	0.139*** (0.015)	0.034*** (0.011)	0.029* (0.015)	0.011 (0.016)	0.031*** (0.011)	0.041*** (0.013)	0.019 (0.014)
Firm Age	− 0.006*** (0.002)	− 0.004*** (0.001)	− 0.006 (0.004)	− 0.002 (0.002)	− 0.006 (0.005)	− 0.005 (0.003)	− 0.007** (0.003)	− 0.001 (0.003)	− 0.013*** (0.005)
Depreciation	0.002 (0.035)	0.104*** (0.020)	− 0.001 (0.046)	− 0.038 (0.032)	0.029 (0.082)	− 0.020 (0.041)	0.008 (0.042)	− 0.066 (0.041)	0.039 (0.050)
Observations	14,429 ^a	6980	7110	7576	6643	7434	6758	7699	6557
Firm fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
R ²	0.676	0.754	0.721	0.698	0.693	0.676	0.708	0.679	0.697

^a We note that the median of share repurchases (reported in Table 1 as 0.000) is 0.0000131. Hence, the subsample which contains only firm-years with repurchase activity is larger than the half of the full sample.

low growth opportunities. In contrast the share repurchases–investment relationship is negative and statistically significant for the high growth opportunities subsample. As in the case of low external constraints, firms that are characterized by low growth do not need to align their share repurchase activity to their investments. Therefore, all else equal, firms with low growth opportunities can fund their comparatively lower financing needs with internal capital without the risk of having to resort to external financing. On the contrary, the opposite holds for high

growth opportunities firms. Such firms reduce share repurchases comparatively more either to secure sufficient investment capital or to avoid costlier external finance. Our findings, also support hypotheses H_{4a} and H_{4b} regarding the effect of cross-shareholdings on the share repurchases–investment relationship. It appears that resistant to change, high cross-shareholdings firms did not exploit the deregulation of share repurchases and integrate this payout mechanism into their investment decision making. One potential explanation is that these firms, often

criticized for their low economic efficiency, use share repurchases as a method to preserve cross-shareholdings rather than improving their financial flexibility. Franks et al. (2018) provide evidence that share repurchases in Japan are used by entrenched management in such firms as a method to preserve high cross-shareholdings. When the firm buys back shares from insiders it can later resell these to other insiders or strategic partners thus maintaining unearned support from cross-shareholders. On the contrary, firms with low company cross-shareholdings whose management lacks the supportive cushion of investor support against inefficient practices exploit the flexibility of share repurchases and coordinate their repurchase activity to investment needs. This observation supports the wide criticism against this distinct feature of Japanese firms which is believed to lead to poor economic efficiency. The fact that share repurchases appear not to be integrated into such firm's decision making in a manner which would have increased their financial flexibility supports this contention.

In comparison to earlier research, our findings largely mirror US and UK based studies which indicate that share repurchases foster investment (Bliss et al., 2015; Chasiotis & Georgantopoulos, 2021; Iyer et al., 2017). It appears that Japanese firms integrate share repurchases into their investment decision in a similar manner to their Anglo-American counterparts. This manner seems to be shaped by the often-cited need for financial flexibility and the presence of financing frictions. However, the fact that this behavior does not seem to hold for inefficient, non-value maximizing firms as proxied by high cross-shareholdings underlines the importance of the specificities of the Japanese market.

Looking at Table 2 results and our control variables the coefficients of cash flows (Cash Flows) and tangibility (Tangibility) have the expected positive sign. This supports the notion that investment is sensitive to levels of internal capital and to firms borrowing capacity and is in line with earlier empirical studies (see Aivazian et al., 2005; Bond & Meghir, 1994; Carpenter & Guariglia, 2008; Fazzari et al., 1988; Iyer et al., 2017). Our measure for growth opportunities (Growth) also displays a consistent and expected positive relationship to investment. Dividends appear to be positively related to investment. In this respect, the behavior of Japanese firms is in stark contrast to their US and UK counterparts where dividend decisions are taken independently from investment decisions (Allen & Michaely, 2003; Baker, Powell, & Veit, 2002; Brav et al., 2005; Chasiotis & Georgantopoulos, 2021; Iyer et al., 2017; Lintner, 1956; Reneboog & Trojanowski, 2011). This divergent behavior was indicated earlier by Kato et al. (2002) who report that Japanese firms appear to increase dividends in the presence of investment activity. This may indicate that dividends are only increased if higher and sustainable profits are expected from new investment. Since our study employs annual data, our findings also support Dewenter and Warther's (1998) who show that Japanese firms adjust dividends more quickly than their US counterparts. Leverage appears to have a negative effect on investment. This has two possible explanations. Higher use of debt can be a discipline mechanism to control potential overinvestment (Aivazian et al., 2005) and hinder investment indirectly by making external capital costlier. Firm size (Size) appears to be positively related to investment suggesting that this variable captures a firm's ability to generate internal funds and easier access to external capital. The firm age (Firm Age) variable appears to have a negative effect on investment in line with the notion that mature firms have lower investment opportunities. Finally, depreciation (Depreciation) generally has the expected positive sign however it is generally insignificant.

4.2. Robustness tests

4.2.1. Endogeneity tests - instrumental variable analysis (2SLS)

In this section, to alleviate possible endogeneity issues in our model that may affect our baseline estimations we utilize two-stage least square (2SLS) estimations. McCabe (1979) argues that investment, dividends and share repurchases are competing uses of funds. This notion suggests endogeneity issues that need to be accounted for. Our set of instruments

is based on Brockman, Khurana, and Martin (2008) and Swift (2018). Accordingly, to instrument share repurchases we use the predicted probability of repurchase activity using a set of predetermined variables¹¹ and the percentage change in the price-earnings ratio of all firms in this sample in the prior year. Results are reported in Table 3. As an additional robustness check,¹² we also use the industry's mean of share repurchases as an instrument following Adhikari and Agrawal (2018). To provide validity in our 2SLS estimations, we use the under-identification test by Kleibergen and Paap (LM statistic) to check if the number of instruments is adequate compared with the number of the endogenous variables. The null hypothesis is that there is under-identification and a p -value lower than 0.05 and 0.1 is required to reject the null hypothesis at the 5% and 10% level, respectively. We use the Hansen over-identification (p -value) test to test for possible correlation between the instruments and the error term. If there is any correlation then the instruments are not treated as acceptable. Under the null hypothesis over identifying restrictions are valid, and a higher value than 0.05 is required to reject the null hypothesis at the 5% level. We check for the instrument's explanatory power by using a weak identification test. In this test, we compare the critical values with those of the Cragg-Donald Wald F-statistic and if any of them is greater than that, then the instruments are weak and do not have explanatory power. All the above statistics show that we performed correctly the 2sls procedure. It is important to note that our instruments are valid according to the Hansen test of instrument validity (Hansen's J-statistic). The joint null hypothesis is that the instruments are valid, and thus, uncorrelated with the error term. We fail to reject the null according to the p -values of the statistic. In the first stage of our regression, we estimate the instruments' impact on the control variables of our baseline model, while in the second stage we regress firm investment on the first-stage residuals including the former control variables. In Table 3 we present the results from the two-stage least squares estimations. After controlling for endogeneity, results are in line with those of Table 3 and thus provide further support for hypotheses $H_1 - H_{4a,b}$.

4.2.2. Endogeneity tests - entropy balancing approach

In this section, as an alternative approach to alleviate the endogeneity concern, we follow recent literature and employ Hainmueller's (2012) entropy balancing approach. We divide our firm-year observations into two groups based on share repurchase activity. Specifically, the treatment group includes observations with repurchase activity whereas the control group includes observations with no repurchase activity. The preprocessing scheme of entropy balancing involves reweighting observations to balance the mean, variance, and skewness of all covariates across the treatment and control groups. Table 4 presents the entropy balance weighting scores and the regression results. We observe that the relationship between share repurchases and investment is still negative and statistically significant after employing the entropy balancing scheme.

4.2.3. Heckman selection model

To explore our findings further, we account for possible selection mechanisms that could drive our results by employing the Heckman two stage selection model (Table 5). Specifically, omitted variables may be a source of endogeneity. Unobservable firm characteristics related to repurchases activity may also affect investment activity. For example, a firm's ability to generate cash flows may stimulate higher payouts and firm investment. In this case, the coefficient on share repurchases may

¹¹ Brockman et al. (2008) use the risk-free rate, the standard deviation of monthly-weighted adjusted returns and lagged values of the firm's cash flows, return on equity, market capitalization, Tobin's Q and the number of repurchases the previous year.

¹² Results are similar to the ones reported in Table 3 and are available from the authors upon request.

Table 3

This table presents the results from two-stage least squares estimations. Robust standard errors are in parentheses. Notes: Variable definitions are provided in [Table A1](#), appendix. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Full sample	Subsamples based on financial constraints, growth opportunities and investment intensity							
		Internal Financing Constraints		External Financing Constraints		Growth opportunities		Cross-Shareholdings	
		Low	High	Low	High	Low	High	Low	High
Repurchases	− 0.135* (0.074)	− 0.003 (0.043)	− 0.364* (0.187)	− 0.000 (0.087)	− 0.779** (0.315)	− 0.115 (0.330)	− 0.220* (0.121)	− 0.206* (0.103)	− 0.080 (0.129)
Dividends	0.291*** (0.043)	− 0.003 (0.118)	1.034** (0.483)	0.499 (0.317)	4.101** (1.909)	5.025** (2.195)	1.866** (0.785)	0.770 (0.498)	− 0.337 (0.552)
Leverage	− 0.011*** (0.002)	− 0.011*** (0.003)	0.001 (0.010)	0.019** (0.008)	0.051 (0.035)	0.074* (0.039)	0.025 (0.017)	0.002 (0.014)	− 0.029** (0.013)
Firm size	0.002*** (0.001)	0.000 (0.001)	0.001 (0.001)	0.002 (0.001)	− 0.003 (0.004)	− 0.002 (0.005)	− 0.002 (0.002)	0.001 (0.003)	0.005** (0.002)
Tangibility	0.139*** (0.003)	0.031*** (0.002)	0.157*** (0.004)	0.177*** (0.005)	0.133*** (0.014)	0.128*** (0.005)	0.163*** (0.005)	0.163*** (0.019)	0.132*** (0.004)
Growth	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.000)	0.001* (0.000)	0.000 (0.001)			0.002** (0.001)	0.001*** (0.000)
Cash Flows	0.024*** (0.004)	0.063*** (0.004)	0.132*** (0.008)	0.016** (0.007)	0.019 (0.020)	− 0.012 (0.010)	0.002 (0.011)	0.022** (0.009)	0.024*** (0.008)
Firm Age	− 0.010*** (0.001)	− 0.002** (0.001)	− 0.013*** (0.003)	− 0.008*** (0.002)	− 0.011* (0.006)	− 0.011*** (0.004)	− 0.016*** (0.003)	− 0.004 (0.004)	− 0.015*** (0.003)
Depreciation	0.040*** (0.013)	0.122*** (0.010)	0.002 (0.023)	0.012 (0.021)	0.006 (0.063)	− 0.052 (0.036)	0.017 (0.024)	− 0.031 (0.069)	0.087*** (0.019)
Observations	23,611	11,600	11,717	10,655	11,516	10,928	10,318	11,697	11,707
Firm fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
Under identification test	0.000	0.000	0.035	0.000	0.000	0.084	0.000	0.000	0.000
Weak identification test	113.71	29.51	27.26	26.57	29.45	22.40	25.57	30.86	22.580
Hansen J statistic	0.170	0.120	0.16	0.910	0.11	0.23	0.15	0.15	0.18

be upward-biased. The Heckman two-stage procedure will allow us to address the possibility of omitted variables that simultaneously affect share repurchases and investment. We use two specifications considering firms selection that may arise by focusing on share repurchases increases (Specification 1) and share repurchases activity (Specification 2). In the main equation results of Specification 1 and Specification 2 we document that when we address for self-selection at 1% increase in *Repurchases* relates to a reduction of *Investment* by 0.19 to 0.065, according to the specification we follow. The former findings indicate that when we account for self-selection the relationship between investment and shares repurchases becomes stronger and remains negative and statistically significant. These findings confirm our baseline results and provide further robustness.

4.2.4. Quantile regression

In this section, we further explore the share repurchases-investment relationship at several points of the investment distribution, other than the conditional mean. The estimated coefficients of share repurchase in several quantiles (q5, q10, q25, q50, q75, q90, q95) reveal the investment-share repurchases relationship to be heterogeneous (see [Table 6](#)). The negative relationship documented in our previous conditional mean estimations is concentrated at the top end of the investment distribution and becomes stronger (the coefficient doubles in size) in successive quantiles (q75, q90, q95). On the contrary, at all quantiles below the 75th, the respective coefficients are comparatively much smaller in size and mostly statistically insignificant. This suggests that when investment needs are low firms do not need to cut back on share repurchases. Similar heterogeneity issues in the payouts/share repurchases - investment relationship is also reported by [Mathur et al.](#)

(2016) and [Chasiotis and Georgantopoulos \(2021\)](#) for US and UK firm samples, respectively. Overall, our investigation has provided robust evidence regarding the share repurchases – investment relationship. So far, our findings suggest that Japanese firms exploit the inherent flexibility of share repurchases and align their share repurchase activity to their investment needs.

4.2.5. Using alternative proxies

[Table 7](#) provides the results from re-estimating our sample using three alternative proxies for measuring financial constraints. Specifically, we use the [Kaplan and Zingales \(1997\)](#), the [Whited and Wu \(2006\)](#) and the [Hadlock & Pierce, \(2010\)](#) indices. Results, mirror the baseline results presented in [Table 2](#) and show clear support for hypotheses $H_2 - H_{4a,b}$. Moreover, results regarding control variables largely resemble our initial results found in [Table 2](#). In [Table 8](#), we confirm the stability of our results through additional robustness checks. We use alternative proxies for certain explanatory variables. More specifically, we use long term debt to total assets to measure leverage (Leverage'). We measure firm size (Firm Size') as the natural logarithm of market capitalization. We use EBITDA to total assets instead of cash flows from operations (Cash Flows'). Finally, Growth is replaced by sales growth (Growth'). Finally, we explore the sensitivity of our results to different specifications of our model in terms of excluding different sets of control variables. The latter results are available from the authors upon request.

5. Discussion

Our findings considering the negative effect of share repurchases on

Table 4

This table documents the results from the entropy balancing approach. Panel A present the mean, variance and skewness between the treated and control groups before and after weighting. Panel B reports the entropy balancing regression estimates. Standard errors in panel B are reported in parentheses. Variable definitions are reported in [Table A1](#), appendix. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Panel A: Entropy Balancing Weighting						
	Treat			Control		
	mean	variance	skewness	mean	variance	skewness
Before: Without weighting						
Dividends	0.010	0.000	2.136	0.009	0.000	2.306
Leverage	0.456	0.036	0.109	0.524	0.040	− 0.131
Firm Size	18.490	2.134	0.482	18.310	2.187	0.397
Tangibility	0.293	0.027	0.559	0.289	0.031	0.492
Growth	1.457	1.810	3.699	1.617	2.655	3.144
Cash Flows	0.072	0.002	0.594	0.070	0.002	0.532
Firm Age	4.000	0.346	− 2.441	3.896	0.445	− 1.744
Depreciation	0.033	0.000	3.172	0.034	0.001	1.595
After: Weighting variables						
Dividends	0.010	0.000	2.136	0.010	0.000	2.093
Leverage	0.456	0.036	0.109	0.456	0.040	0.159
Firm size	18.490	2.134	0.482	18.490	2.133	0.387
Tangibility	0.293	0.027	0.559	0.293	0.029	0.567
Growth	1.457	1.810	3.699	1.457	1.876	3.468
Cash Flows	0.072	0.002	0.594	0.072	0.002	0.443
Firm Age	4.000	0.346	− 2.441	4.000	0.308	− 1.886
Depreciation	0.033	0.000	3.172	0.033	0.000	1.451

Panel B: This table presents results from the entropy balance method.		
	Full sample	Restricted sample (share repurchases>0)
Repurchases	− 0.083*** (0.015)	− 0.071*** (0.017)
Dividends	− 0.087*** (0.022)	− 0.041 (0.029)
Leverage	− 0.006*** (0.001)	− 0.005*** (0.001)
Firm size	0.001*** (0.000)	0.002*** (0.000)
Tangibility	0.072*** (0.001)	0.069*** (0.001)
Growth	0.001*** (0.000)	0.001*** (0.000)
Cash flows	0.094*** (0.004)	0.103*** (0.006)
Firm Age	0.000 (0.000)	0.001*** (0.000)
Depreciation	0.493*** (0.008)	0.465*** (0.010)
Observations	23,637	14,772
R ²	0.455	0.455
Industry fixed effects	YES	YES
Year fixed effects	YES	YES

a firm's investment policies suggest that the theoretical arguments behind the positive impact of the integration of share repurchases into Japanese firms' decision making are the dominant ones in our sample. These findings provide support for both the pecking order theory of financing and the flexibility hypothesis of share repurchases. Specifically, our results show an inverse relationship between share repurchases and investments. Consistent with our expectations, this suggests that Japanese firms curb share repurchases in the presence of investment needs and vice versa. As [Myers \(1984\)](#) argues, firms prefer internal over external financing, since the latter is costlier due to information asymmetries. To mobilize internal finance, firms are expected to utilize their free cash flows and even adjust their payout policies to

fund investment opportunities. However, adjusting dividends downwards may not be possible due to their rigid nature. Therefore, firms exploit the inherent flexibility of share repurchases to avoid resorting to costly external finance.

Our findings considering the flexibility of share repurchases under internal and external financial constraints have also economic significance. The importance of the flexibility of share repurchases is revealed in the presence of a high level of financial constraints. When there is sufficient internal capital, firms do not coordinate the investment and the repurchase decision. On the contrary, companies with inadequate internal funds, tend to reduce shares repurchase to secure investment capital. Accordingly, the link between share repurchases and investment

Table 5

This table presents the Heckman two-stage procedure. The dependent variable is firm's investment whereas the key independent variable is the predicted value of the probability of repurchases increase (specification 1) and the predicted value of repurchases activity (specification 2). In both specifications, repurchases increase and repurchases activity both are dummy variables that take the value of one if firms increase the amount of repurchases or engage in repurchase activity. Variable definitions are provided in Table A1, Appendix. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors are reported in the parenthesis. All specifications include Industry and year fixed effects.

	Specification 1		Specification 2	
	Main equation	Selection equation	Main equation	Selection equation
Repurchases	– 0.185*** (0.042)		– 0.065*** (0.017)	
Dividends	– 0.043 (0.039)	– 0.108 (1.213)	– 0.029 (0.030)	6.713*** (1.225)
Leverage	– 0.009*** (0.003)	– 0.673*** (0.055)	– 0.006*** (0.002)	– 1.242*** (0.055)
Firm size	0.002*** (0.000)	0.034*** (0.007)	0.002*** (0.000)	0.090*** (0.007)
Tangibility	0.064*** (0.002)		0.067*** (0.001)	
Growth	0.001*** (0.000)	0.007 (0.007)	0.001*** (0.000)	– 0.026*** (0.007)
Cash flows	0.091*** (0.008)	0.397* (0.226)	0.098*** (0.006)	– 1.005*** (0.223)
Firm age	0.002*** (0.001)		0.001*** (0.000)	
Depreciation	0.520*** (0.017)		0.477*** (0.011)	
Cross-shareholdings		– 0.596*** (0.074)		– 1.060*** (0.073)
Stock options		29.717 (26.541)		– 30.623 (26.359)
Stock returns		0.200*** (0.029)		0.101*** (0.028)
Ownership concentration		0.125* (0.073)		0.138* (0.073)
lambda	0.011*** (0.004)		0.003 (0.002)	lambda
Constant	– 0.057*** (0.009)	– 0.754*** (0.138)	– 0.046*** (0.006)	– 0.595*** (0.138)
Observations	21,645	21,645	21,634	21,634
Year fixed effects	YES	YES	YES	YES
Industry fixed effects	YES	YES	YES	YES

relationships is driven by the firm's ability to raise external capital. When access to external funding is limited share repurchases are secondary to investment. On the contrary, investment and repurchase decisions are independent when access to external capital is unproblematic. Moreover, growth opportunities determine the investment-repurchases nexus. This implies that in the presence of significant prospects firms coordinate share repurchases to investment to avoid the higher costs of external finance.

Equally relevant is the level of investment intensity in shaping the investment-repurchases nexus. High investment levels correspond to higher capital needs thus forcing firms to divert funds from share repurchases towards investment and avoid costly external financing. The results underline the relevance of estimating the integration of share repurchases at different investment levels, representing different capital needs.

The investment-share repurchases relationship is also influenced by cross-shareholdings, a distinct trait of the Japanese market. This unique characteristic provides no incentive for efficient use of capital as management enjoys undeserved support by cross-holders. Thus, firms with

high levels of cross-shareholdings do not integrate share repurchases into their decision-making at the expense of their financial flexibility.

This study fails to provide evidence against the use of share repurchases. The results do not suggest that firms are willing to trade off investment for share repurchases. The fact that share repurchases are curtailed in the presence of financial constraints, growth opportunities, and investment intensity suggest that the repurchase decision is secondary to investment confirming managerial views as reported by Brav et al. (2015). If the repurchase decision was made irrespective of the investment decision no relationship between share repurchases and investment would have been observed.

6. Concluding remarks and policy implications

This study is motivated by the marked increase in share repurchase activity in Japan following their deregulation in the 1990s. It joins the ongoing debate regarding this payout mechanism's dubious effect on investment predominantly originating from their increasing use in the US. Proponents of share repurchases consider them an indispensable tool for management, since contrary to 'sticky' dividends, share repurchases can be unreluctantly curbed to strengthen liquidity and/or to procure investment capital (Bliss et al., 2015; Brav et al., 2005; Iyer et al., 2017). Critics of share repurchases argue that share repurchases are associated with distorted incentives and this may lead to myopic business practices where management trades off investment for share repurchases thus hampering long-term growth (Almeida et al., 2016; Kahle, 2002; Young & Yang, 2011). Thus, this study explored how this payout mechanism has been integrated into Japanese firms' financial decision making, namely investment.

The baseline model suggests an inverse relationship between share repurchases and investment. This is in line with survey-based evidence that managers consider investment needs first and adjust share repurchases accordingly (i.e., share repurchases are curtailed in the presence of investment needs and vice versa). Further analysis on appropriate subsamples supports this contention as the inverse relationship between share repurchases and investment is more profound in the presence of financial constraints, growth opportunities. This relationship holds after investigating and correcting for self-selection, endogeneity and the use of alternative proxies for several regressors. Finally, after employing quantile regressions as a further robustness test, we shed light on the share repurchases -investment relationship at different points of the investment distribution. Relevant results show that this relationship is heterogeneous and concentrated at the higher end of the investment distribution (75th, 90th, 95th) becoming stronger at consecutive quantiles. Ceteris paribus, firms with higher investment activity curb repurchases comparatively more. Overall, the weight of our evidence shows that that Japanese firms exploit the inherently flexible nature of share repurchase and adjust share repurchases according to their investment needs. This way integration of share repurchases into investment decision making is consistent with the notion that share repurchases foster rather than impede investment.

The findings have direct policy implications as they show that the deregulation of share repurchases granted Japanese firms a valuable distribution mechanism that increases their financial flexibility. This is especially important since reforms under Abenomics targeted the poor economic efficiency of Japanese firms. However, it is important to note that several firms, characterized by high company-cross shareholdings, appears to resist policy reforms and do not improve their economic efficiency. This supports plans of the Tokyo Stock Exchange to provide further incentives to Japanese firms to reduce their company cross-shareholdings. Thus, it would be interesting for further research to monitor firm response to the implementation of these new regulations.

Table 6

This table reports the unconditional quantile regression with fixed effects results at the 5th, 10th, 25th, 50th, 75th, 90th, 95th quantiles, respectively (Q5-Q95). Variable definitions are provided in Table A1, appendix. Cluster-robust standard errors at the firm level are reported in parentheses. $p < 0.01$, $** p < 0.05$, $* p < 0.1$.

	Q5	Q10	Q25	Q50	Q75	Q90	Q95
Repurchases	− 0.007 (0.005)	− 0.009* (0.005)	0.002 (0.007)	0.000 (0.009)	− 0.024* (0.014)	− 0.043* (0.025)	− 0.095* (0.049)
Dividend	0.008 (0.019)	− 0.000 (0.019)	0.023 (0.029)	0.053 (0.033)	0.153** (0.061)	0.156* (0.091)	0.472** (0.186)
Leverage	0.001 (0.001)	0.001 (0.002)	− 0.008*** (0.002)	− 0.016*** (0.004)	− 0.022*** (0.006)	− 0.018* (0.010)	− 0.040** (0.018)
Firm size	− 0.001 (0.000)	− 0.001 (0.001)	0.001 (0.001)	0.002** (0.001)	0.004*** (0.002)	0.007*** (0.003)	0.010 (0.006)
Tangibility	0.006** (0.003)	0.013*** (0.003)	0.036*** (0.004)	0.094*** (0.006)	0.176*** (0.011)	0.304*** (0.019)	0.486*** (0.038)
Growth	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000*** (0.000)	0.001** (0.000)	0.001*** (0.000)	0.002*** (0.001)
Cash Flows	0.007** (0.003)	0.014*** (0.003)	0.023*** (0.004)	0.025*** (0.006)	0.019** (0.009)	0.033* (0.017)	0.047 (0.033)
Firm Age	− 0.001 (0.001)	− 0.002 (0.001)	− 0.003 (0.002)	− 0.005** (0.003)	− 0.010** (0.004)	− 0.021*** (0.007)	− 0.043*** (0.014)
Depreciation	0.022** (0.009)	0.031*** (0.011)	0.041*** (0.014)	0.065*** (0.021)	0.105** (0.041)	0.012 (0.084)	− 0.063 (0.204)
Constant	0.016* (0.009)	0.021* (0.012)	− 0.002 (0.015)	− 0.017 (0.020)	− 0.041 (0.031)	− 0.070 (0.053)	− 0.057 (0.123)
Obs.	23,740	23,740	23,740	23,740	23,740	23,740	23,740
Firm fixed effects	YES	YES	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES	YES	YES
R ²	0.005	0.012	0.030	0.055	0.068	0.060	0.046

Table 7

Results from estimating Eq. (1) with high dimensional firm and year fixed effects and alternative measures of financial constraints. Robust standard errors clustered at the firm level are reported in parentheses. Variable definitions are reported in Table A1, appendix. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

		Subsamples based on alternative measurements of financial constraints					
		Kaplan and Zingales index		Whited-Wu index		Hadlock and Pierce index	
		Low	High	Low	High	Low	High
Repurchases	− 0.037** (0.015)	− 0.026 (0.020)	− 0.040* (0.021)	− 0.029 (0.024)	− 0.040** (0.019)	− 0.024 (0.024)	− 0.037 (0.027)
Dividends	0.186*** (0.050)	0.177** (0.082)	0.225*** (0.058)	0.258*** (0.067)	0.174** (0.073)	0.075 (0.054)	0.708*** (0.152)
Leverage	− 0.014*** (0.003)	− 0.017*** (0.005)	− 0.011** (0.005)	− 0.011** (0.005)	− 0.016*** (0.005)	0.001 (0.007)	− 0.018*** (0.007)
Firm size	0.003** (0.001)	0.003* (0.002)	0.003* (0.002)	0.008*** (0.002)	0.001 (0.002)	0.006** (0.002)	− 0.000 (0.002)
Tangibility	0.138*** (0.007)	0.142*** (0.010)	0.142*** (0.010)	0.138*** (0.010)	0.141*** (0.010)	0.175*** (0.013)	0.154*** (0.011)
Growth	0.002*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002* (0.001)	0.001*** (0.000)
Cash Flows	0.026*** (0.007)	0.021** (0.010)	0.021** (0.009)	− 0.001 (0.009)	0.044*** (0.011)	0.026* (0.013)	0.032** (0.014)
Firm age	− 0.010*** (0.003)	− 0.003 (0.002)	− 0.023*** (0.007)	− 0.040*** (0.011)	− 0.005** (0.002)	− 0.007** (0.003)	− 0.019*** (0.007)
Depreciation	0.042 (0.032)	0.056 (0.047)	− 0.002 (0.029)	− 0.039 (0.029)	0.092* (0.050)	0.101 (0.079)	− 0.057 (0.070)
Constant	− 0.012 (0.021)	− 0.048 (0.032)	0.026 (0.035)	0.024 (0.049)	− 0.008 (0.031)	− 0.097** (0.043)	0.072 (0.055)
Obs.	23,740	11,759	11,764	11,753	11,909	11,854	11,626
Firm fixed effects	YES	YES	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES	YES	YES
R ²	0.666	0.720	0.637	0.640	0.697	0.673	0.636

Table 8

Results from estimating Eq. (1) with high dimensional firm and year fixed effects and alternative proxies for a set of control variables, denoted by ('). The full sample is split into subsamples based on financing constraints, growth opportunities and cross-shareholdings. Robust standard errors clustered at the firm level are reported in parentheses. Variable definitions are reported in Table A1, appendix. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Full sample	Subsamples based on financial constraints, growth opportunities and company cross– shareholdings							
		Internal Financing Constraints		External Financing Constraints		Growth opportunities		Company cross– shareholdings	
		Low	High	Low	High	Low	High	Low	High
Repurchases	– 0.038*** (0.015)	– 0.015* (0.009)	– 0.080*** (0.026)	– 0.034* (0.020)	– 0.048** (0.020)	– 0.021 (0.020)	– 0.054*** (0.019)	– 0.051** (0.020)	– 0.021 (0.022)
Dividends	0.217*** (0.050)	0.037 (0.028)	0.407*** (0.088)	0.281*** (0.077)	0.298*** (0.056)	0.239*** (0.066)	0.188*** (0.065)	0.260*** (0.068)	0.225*** (0.069)
Leverage'	0.008* (0.005)	– 0.013*** (0.003)	0.018*** (0.006)	0.001 (0.005)	0.016** (0.008)	0.008 (0.005)	0.009 (0.007)	0.011* (0.006)	0.007 (0.007)
Firm size'	0.003*** (0.001)	0.002*** (0.000)	0.004*** (0.001)			0.004*** (0.001)	0.002*** (0.001)	0.004*** (0.001)	0.003*** (0.001)
Tangibility	0.138*** (0.007)	0.032*** (0.004)	0.155*** (0.008)	0.134*** (0.011)	0.147*** (0.010)	0.120*** (0.008)	0.160*** (0.009)	0.162*** (0.010)	0.130*** (0.010)
Growth'	0.004*** (0.002)	0.000 (0.001)	0.005** (0.002)	0.001 (0.002)	0.007*** (0.002)			0.003 (0.002)	0.006** (0.002)
Cash Flows'	0.048*** (0.006)	0.050*** (0.004)	0.106*** (0.009)	0.054*** (0.007)	0.058*** (0.009)	0.029*** (0.007)	0.066*** (0.009)	0.046*** (0.008)	0.043*** (0.009)
Firm age	– 0.009*** (0.002)	– 0.003** (0.001)	– 0.014*** (0.004)	– 0.002 (0.002)	– 0.023*** (0.007)	– 0.004* (0.003)	– 0.012*** (0.004)	– 0.003 (0.003)	– 0.016*** (0.004)
Depreciation	0.032 (0.032)	0.125*** (0.019)	0.018 (0.044)	0.035 (0.048)	0.006 (0.035)	0.084* (0.048)	– 0.001 (0.035)	– 0.033 (0.040)	0.073 (0.045)
Constant	– 0.032** (0.014)	– 0.014* (0.008)	– 0.032 (0.021)	– 0.002 (0.009)	0.073*** (0.026)	– 0.071*** (0.017)	– 0.009 (0.019)	– 0.071*** (0.018)	– 0.004 (0.023)
Obs.	23,740	11,648	11,810	12,267	11,422	11,752	11,751	11,791	11,751
Firm fixed effects	23,740	11,648	11,810	12,267	11,422	11,752	11,751	11,791	11,751
Year fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
R ²	YES	YES	YES	YES	YES	YES	YES	YES	YES

Appendix

Table A1

Variable definitions

Variable (Name)	Worldscope items
Investment (<i>Investment</i>): Capital expenditures scaled by the book value of total assets	WC04601, WC02999
Share repurchases (<i>Repurchases</i>): Purchases of common and preferred stock ^a to book value of total assets.	WC04751, WC02999.
Cash dividends paid (<i>Dividends</i>): scaled by the book value of total assets	WC04551, WC02999.
Cash flows (<i>Cash Flows</i>): Cash flows from operations scaled by the book value of total assets.	WC04201, WC02999.
Cash flows (<i>Cash Flows</i>): EBITDA scaled by the book value of total assets.	WC018198, WC02999.
Leverage (<i>Leverage</i>): Total debt scaled by the book value of total assets	WC03255, WC02999,
Leverage (<i>Leverage</i>): Long-term debt scaled by the book value of total assets	WC03251, WC02999.
Growth opportunities (<i>Growth</i>): Market value of equity to book value of equity	MTBV
Growth opportunities (<i>Growth</i>): (Sales growth _t - Sales growth _{t-1}) / Sales growth _{t-1}	WC08631.
Tangibility (<i>Tangibility</i>): Property, plant and equipment scaled by the book value of total assets	WC02501, WC02999.
Firm Size (<i>Firm Size</i>): Natural logarithm of the book value of total assets.	WC02999.
Firm Size (<i>Firm Size</i>): Natural logarithm of the market capitalization.	WC08001.
Firm age (<i>Firm age</i>): Natural logarithm of a firm's age.	WC18273.
Depreciation (<i>Depreciation</i>): Depreciation scaled by the book value of total assets	W01148, WC02999.
Stock options (<i>Stock Options</i>): Proceeds from stock options to book value of total assets.	WC04301, WC02999.
Ownership concentration (<i>Ownership Concentration</i>): Closely held shares (%)	WC08021
Company cross-shareholdings (<i>Cross-shareholdings</i>): Total holdings by one company in another (%).	NOSHCO
Stock returns (<i>Stock returns</i>): Stock's total return index from Datastream (<i>RI</i>) as $\ln(RI_t / RI_{t-1})$	RI.
Internal Financial constraints: (Net cashflow from operations minus Capex) scaled by Capex	WC04201, WC04601, WC04601.
External Financial constraints: Interest Expenses scaled by total liabilities.	WC01075, WC03251.
Hadlock and Pierce (2010) index: $= -0.737SIZE + 0.043SIZE^2 - 0.040AGE$, where, SIZE: is the logarithm of total assets AGE: is the Firm Age variable	WC18273, WC02999
Whited and Wu (2006) index $= -0.091CF - 0.062DD + 0.021LEV - 0.44LNTA + 0.102ISG - 0.035SG$, where, CF: is the Cash Flows' variable DD: is a dummy variable, which takes the value of 1 if a firm pays dividends and zero otherwise. LEV: is the Leverage variable LNTA: is the Firm Size variable ISG: is the firm's industry sales growth. Industry is defined as the 3-digit industry sic-code SG: is the Growth' variable	WC018198, WC02999, WC04551, WC03255, WC08631, WC01001
Kaplan and Zingales (1997) index: $0.238Q - 1.002CF + 3.139LEVR - 39.368DIV - 1.315CASH$, where, Q: is the market to book ratio (variable Growth)?? CF: is the Cash Flows' variable LEVR: is the Leverage' variable	WC018198, WC02999, WC04551, WC03251, WC01001,

(continued on next page)

Table A1 (continued)

Variable (Name)	Worldscope items
DIV: is the dividends variable	WC02005,
CASH: is the firm's cash and cash equivalents dividend by the book value of total assets	MTBV
^a "Purchases of common and preferred stock" is considered as the most accurate measurement of actual share repurchases (Banyi, Dyl, & Kahle, 2008).	

Table A2

Pairwise correlations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ Indicate two tailed statistical significance. Variable definitions are reported in Table A1, appendix.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) Investment	1.000									
(2) Repurchases	− 0.03***	1.000								
(3) Dividends	− 0.03***	0.27***	1.000							
(4) Leverage	0.03***	− 0.17***	− 0.49***	1.000						
(5) Firm size	0.14***	− 0.02***	− 0.17***	0.26***	1.000					
(6) Tangibility	0.52***	− 0.10***	− 0.23***	0.20***	0.18***	1.000				
(7) Growth	0.05***	0.11***	0.28***	− 0.02***	− 0.17***	− 0.19***	1.000			
(8) Cash flows	0.20***	0.13***	0.40***	− 0.31***	− 0.10***	0.01**	0.43***	1.000		
(9) Firm age	0.02**	− 0.08***	− 0.16***	0.09***	0.26***	0.17***	− 0.34***	− 0.24***	1.00	
(10) Depreciation	0.56***	0.01*	0.01**	0.02***	0.12***	0.42***	0.05***	0.38***	− 0.02***	1.00

Table A3

Variance inflation factors. Variable definitions are reported in Table A1, appendix.

Variable	Dividends	Leverage	Cash Flows	Growth	Tangibility	Firm Size	Repurchases	Firm age	Depreciation
VIF	1.60	1.53	1.77	1.47	1.38	1.17	1.09	1.20	1.49

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